Biochimica et Biophysica Acta 1814 (2011) 36–45

Contents lists available at ScienceDirect

Biochimica et Biophysica Acta

journal homepage: www.elsevier.com/locate/bbapap

Review

P450s in plant–insect interactions

Mary A. Schuler *

Departments of Cell and Developmental Biology, Biochemistry and Plant Biology University of Illinois, Urbana, Illinois 61801, USA

ARTICLE INFO

Article history:
Received 14 May 2010
Received in revised form 15 September 2010
Accepted 16 September 2010
Available online 29 September 2010

Keywords:
Cytochrome P450 monooxygenase
Plant–insect interactions
Plant metabolites
Insect detoxification system

ABSTRACT

Cytochrome P450 monooxygenases (P450s) are integral in defining the relationships between plants and insects. Secondary metabolites produced in plants for protection against insects and other organisms are synthesized via pathways that include P450s in many different families and subfamilies. Survival of insects in the presence of toxic secondary metabolites depends on their metabolism by more limited groups of P450s. Examples of functionally characterized plant and insect P450s known to be involved in these interactions are discussed in terms of their diversities, reactivities and regulators. These and future examples, which will be uncovered as the fields of plant biology and entomology converge on this interesting area, provide much insight into the array of plant metabolites that are mainline defenses against insects, the range of insect monooxygenases that inactivate these compounds and the evolutionary processes occurring as these organisms wage daily battles with one another. Molecular perspectives on these interactions will provide the scientific community with information critical for genetic manipulation of these organisms aimed at enhancing plant resistance to insects and eliminating insect resistance to natural plant toxins and synthetic insecticides.

1. Introduction

P450s have long been recognized as exceptionally unusual in their ability to mediate the complex insertions of molecular oxygen into different substrates. It is only now becoming evident through genomic sequencing efforts that the number and range of P450s within an individual organism reflect, to some extent, its lifestyle and its mode of defense against toxins in its environment. Plants and herbivorous insects provide especially interesting examples of this relationship as they wage a continuing battle with one another — with one being a stationary food source and the other being an exploratory consumer of plant components. Success of the first (plants) depends on production of an array of toxic defense compounds fending off the second (insects). Success of the second depends on inactivation and/or evasion of the most detrimental toxins.

As plants and insects have evolved in their ecological interactions and biochemical systems, P450 gene numbers have increased much more dramatically in plant genomes than in insect and vertebrate genomes. Examples of the many plant pathways that have evolved for producing ecologically important secondary metabolites as defenses against insect consumers are detailed later in this article but even these do not begin to do justice to complexities of plant secondary products. The evolution of these complex pathways has often necessitated that stereo- and regio-specific P450s be recruited for the biochemical processing and production of oxygenated intermediates. Frequently, expansions in these biochemical capacities have arisen from cycles of duplication and divergence of reiterated gene copies and resulted in many substrate-selective P450s in each pathway and correspondingly large numbers of genes in plant genomes. Less frequently, the evolution of these pathways has caused substrate-processive P450s mediating multiple conversions on the same substrate to be included in each pathway. Examples of the insect systems that have evolved to counter the toxic effects of these plant metabolites indicate that evolution of detoxicative enzymes has often occurred by selective mutation of catalytic sites in existing or recently duplicated P450s (which alters substrate range) and/or by substantial mutation of promoter sequences (which enhances expression) and resulted in the maintenance of smaller numbers of genes in insect genomes. Taking into account these different molecular modes of adapting to ecological pressures, this review attempts to highlight our current knowledge of P450s relevant to plant–insect interactions and leaves to other reviews those involved in plant-fungal and plant-bacterial interactions. Recent general reviews on plant P450s include Werck-Reichhart et al. [1], Nielsen and Moller [2], Schuler et al. [3] and Mizutani and Ohta [4]. Recent general reviews on insect P450s include Feyereisen [5], Li et al. [6] and Despres et al. [7]. The diversity of plant secondary products is extensively summarized in Seigler [8] and Wink [9].

2. Plant P450s

2.1. Biosynthetic pathways

With plant P450s generally mediating the same sorts of aliphatic hydroxylations, aromatic hydroxylations, epoxidations, isomerizations, dimerizations, carbon–carbon cleavages, carboxylations and other reactions commonly attributed to mammalian P450s, many

* Department of Cell and Developmental Biology, 190 ERML, 1201 W. Gregory Drive, University of Illinois, Urbana, IL 61801, USA. Tel.: +1 217 333 8784; fax: +1 217 244 1336.
E-mail address: maryschu@illinois.edu.

1570-9639/$ – see front matter © 2010 Published by Elsevier B.V.
doi:10.1016/j.bbapap.2010.09.012

have been recruited to highly specialized functions as plants have evolved new pathways for the production of defense toxins. The most conservative estimates suggest that the 300,000 plant species existing in the world collectively generate over 200,000 natural products [10]. Acknowledging that these numbers are likely to increase as metabolomic profilings reveal the complexities of individual plant pathways, it is clear that P450s have key and often rate-limiting roles in the production of plant allelochemicals.

Among these, terpenoids represent one of the largest classes of plant defense compounds (more than 40,000 structures) that includes toxic and repellent ecological defense compounds (*e.g.*, limonene, perillyl alcohol, citral, myrcene, and pinenes) [11,12]. Alkaloids represent another large and extremely diverse class of plant defense compounds (more than 12,000 structures) that include the pharmaceutically relevant isoquinoline and benzylisoquinoline alkaloids (berbamunine, tubocurarine, morphine, and codeine), monoterpene indole alkaloids (vinblastine, vincristine, quinine, and strychnine), tropane and nicotine alkaloids (nicotine, scopolamine, and atropine), purine alkaloids (caffeine) and alkaloid esters (homoharringtonine) [13–15].

Smaller classes of plant defense compounds also exist. Among the most toxic of these, furanocoumarins (more than 200 structures) are derived by attachment of a furan ring to coumarin in either a linear orientation (xanthotoxin, bergapten, and psoralen) or an angular orientation (angelicin and sphondin) (Fig. 1A) [16–18]. Like their structural furanochromone (visnagin and khellin) relatives, furanocoumarins are phototoxic to a variety of organisms and cell types as a result of their ability to crosslink DNA [18,19]. Glucosinolates (more than 120 structures) are thioglucosides derived from methionine (aliphatic glucosinolates), tryptophan (indole glucosinolates) and phenylalanine (benzylglucosinolates) (Fig. 2) [20,21]. Piperamides (piperidine, piperine, and pipercide), which serve as natural insecticides, are derived from the linkage of amides and methylenedioxyphenyl (MDP) moieties [22]. Benzoxazinoids (DIMBOA) are hydroxamic acids derived from indole [23]. Cyanogenic glucosides (dhurrin) are derived from tyrosine and converted to toxic cyanide by β-glucosidases and hydroxyl nitrile lyases released in insect-damaged plants [24,25].

In addition to these chemically distinct categories, many plant allelochemicals are derived from the phenylpropanoid pathway. Among these, the simplest MDP compounds (myristicin, sesamin, and safrole) are dimeric phenylpropanoids that, like the synthetic piperonyl butoxide (PBO), interfere with P450 activities in a wide range of organisms including insects and mammals [26,27]. Flavonoids (flavone, quercetin, and genistein) are monomeric phenylpropanoids that can also interfere with P450 activities and, at some concentrations and cellular conditions, can act as antioxidants [28–31]. Contrasting with the detrimental effects of these previously mentioned phenylpropanoids, anthocyanins derived from another branch in the phenylpropanoid pathway serve as floral pigments and attractants to insect pollinators [30,32]. Stilbenes (resveratrol, piceatannol, and viniferins) derived from yet another branch have been reported to enhance longevity in *Drosophila* and other metazoans [33,34].

Because insects constantly subjected to these types of defense toxins readily develop resistance against one or more of them, plants contain suites of allelochemicals that can be used in their chemical defense. Some plants contain a wide array of moderately toxic defense compounds while others contain only a small number of highly toxic defense compounds. Furanocoumarins, which occur with various structures in the Apiaceae, provide good examples of the effectiveness of combinatorial mixtures: tested individually, many have detrimental effects on insect growth and development and, tested in mixtures, their chemical combinations are significantly more toxic [35].

2.2. Gene counts and organizations

P450 gene duplications and divergences occurring in the evolutionary process have resulted in astounding numbers of P450 genes in higher plants. In sequenced plant genomes, the P450 gene superfamily includes 245 full-length genes in *Arabidopsis thaliana* (mouse ear's cress), 332 in *Oryza sativa* (rice), 315 in *Vitis vinifera* (grape), 310 in *Populus trichocarpa* (poplar) and 142 in *Carica papaya* (papaya) [1,36–41]. Evolutionary tours of these five plant genomes typically show long clusters of genes within the same P450 subfamily with those in the same clade and related P450 families usually containing introns at common positions [37–39]. These clusters are frequently interspersed with more divergent genes from other P450 subfamilies, as seen in the CYP2ABFGST gene clusters in mammals. Acknowledging that these statements tend to oversimplify the relationships of plant P450 genes, it is the exceptions to them that highlight the high degree of genetic rearrangement in plant genomes and the complexity of P450 evolution in plants. Exceptions occurring in *Arabidopsis* include the CYP71B subfamily (which contains 35 members tandemly arrayed at eight locations among all five of its chromosomes), the large CYP71 and CYP705 families (whose members contain from 0 to 7 introns) and the small CYP96A subfamily (whose members contain 0–3 introns) [3,36].

2.3. P450s in biosynthetic pathways

Among the many plant P450s, relatively few families and subfamilies have been maintained in all sequenced plant genomes. Families existing in *Physcomitrella patens* (moss) and the five higher plants include CYP51 mediating obtusifol 14α-demethylation, CYP74 mediating oxylipin synthesis (jasmonic acid and methyl jasmonate) and catabolism (C_6-volatiles), CYP86, CYP94, CYP703 and CYP704 mediating various fatty acid hydroxylations, CYP97 mediating various carotenoid hydroxylations and CYP701 mediating multiple conversions in gibberellin synthesis [4,39]. Subfamilies existing in these six genomes include CYP73A mediating cinnamic acid hydroxylation in early phenylpropanoid synthesis, CYP78A mediating an unspecified conversion controlling organ development, CYP98A mediating shikimate hydroxylation in the lignin branch of phenylpropanoid metabolism and CYP710A mediating sterol C22-desaturations. Without moss included in these comparisons, families existing in the five higher plants include, CYP85 and CYP90 mediating multiple steps in brassinosteroid synthesis. Subfamilies existing in these higher plants include CYP75B mediating flavonoid hydroxylation, CYP84A mediating ferulate hydroxylation, CYP88A mediating additional conversions in gibberellin synthesis, CYP77A mediating fatty acid hydroxylations, CYP707A mediating abscisic acid (sesquiterpene) catabolism, CYP711A mediating strictolactone (terpenoid lactone) synthesis, CYP734A mediating brassiosteroid catabolism, CYP735A mediating cytokinin catabolism and CYP72A, CYP77B and CYP87A with as-yet-undefined functions [4,39].

Apart from these families and subfamilies that are conserved for common functions in plant development and environmental response, the P450 subfamilies involved in allelochemical production occur in more limited groups of plants because of differences in the range of their secondary metabolites, the substrate specificities of their monooxygenases, the existence of alternate enzymes mediating similar reactions and, sometimes, differences in their physiological processes. In the Brassicaceae such as *Arabidopsis thaliana*, aliphatic glucosinolate synthesis (Fig. 2A) is mediated by CYP79F1 and CYP79F2, each of which has distinct functions in the conversion of short- and long-chain methionine derivatives to oximes [42–44], prior to their modification by CYP83A1 [45–47]. Indole glucosinolate synthesis (Fig. 2B) is mediated by CYP79B2 and CYP79B3, each of which functions in the conversion of tryptophan derivatives to indole-3-acetyldoxime [48,49], prior to its oxidation by CYP83B1 [45,47,50]. Benzylglucosinolate synthesis is mediated by CYP79A2, which mediates the conversion of phenylalanine to oxime [51], prior to its oxidation by CYP83B1, the convergence point for benzylglucosinolate and indole glucosinolate syntheses.

In *Sorghum bicolor* (sorghum), synthesis of the cyanogenic glucoside dhurrin is mediated by the sequential N-hydroxylation of tyrosine,

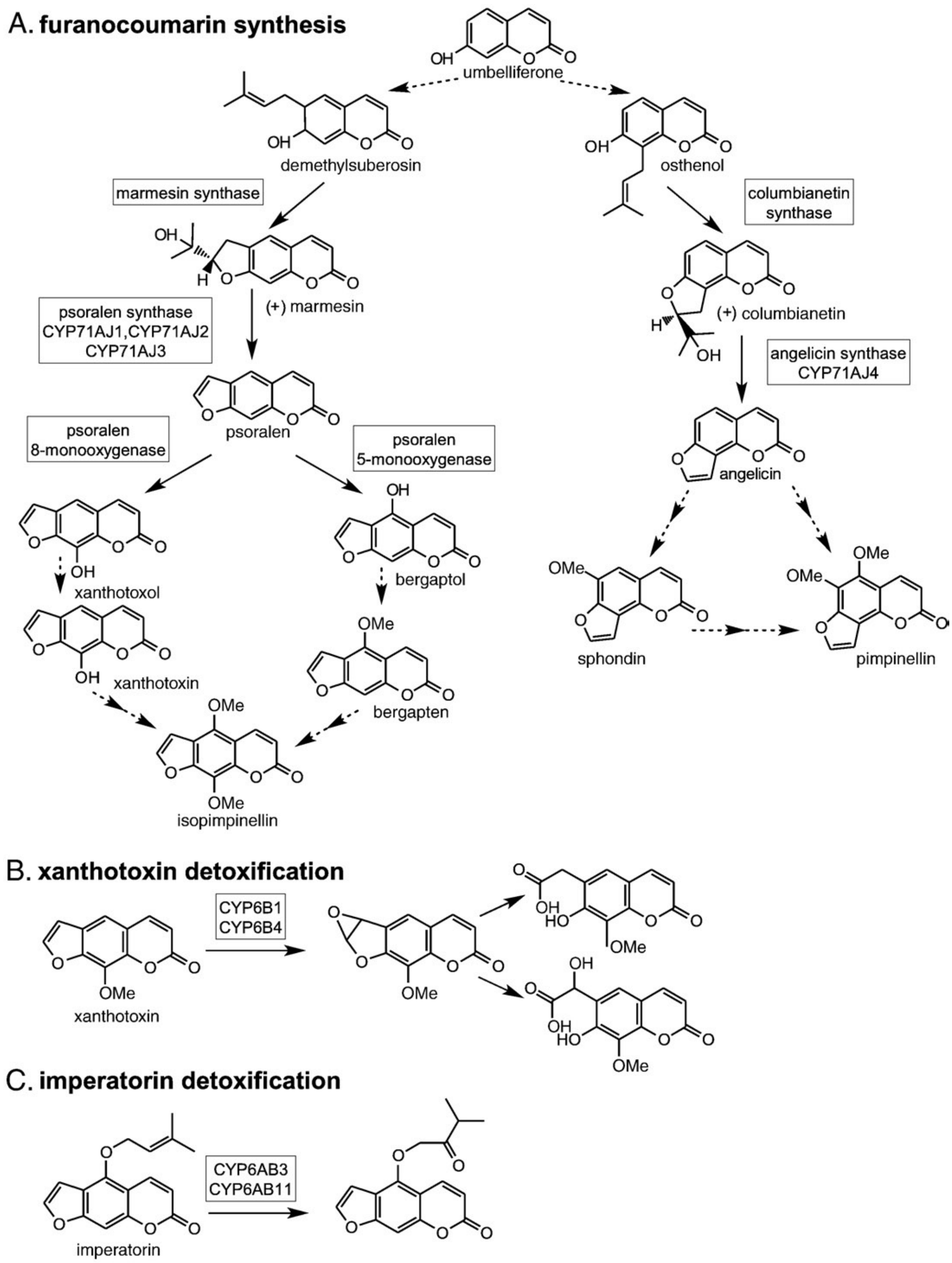

Fig. 1. Furanocoumarin synthesis and degradation. (A) The synthetic pathway for linear (left branch) and angular (right branch) furanocoumarins occurring in some umbelliferous plants is shown with boxes designating known P450 steps. (B) P450 detoxification of the linear furanocoumarin xanthotoxin occurring in some lepidopteran insects (*Papilio polyxenes*, *P. glaucus*, *Helicoverpa zea* and others) occurs via epoxidation on the furan ring and subsequent hydroxylation. (C) P450 detoxification of the linear furanocoumarin imperatorin occurring in some lepidopteran insects (*Depressaria pastinacella*, *Amyelois transitella*) occurs via epoxidation of the extended isoprenyl side chain.

dehydration and decarboxylation to *p*-hydroxyphenyl aldoxime by the multifunctional CYP79A1 [52,53]. This intermediate is subsequently dehydrated and hydroxylated to cyanohydrin *p*-hydroxymandelonitrile by the multifunctional CYP71E1 [54]. Orthologues of CYP79A1, which have been identified in a number of cyanogenic dicots and monocots [55–58], include CYP79D subfamily members in *Manihot esculenta* (cassava) and *Lotus japonicus* (trefoil) that are involved in the synthesis of linamarin and lotaustralin (cyanoalk(en)yl glycosides) from valine and isoleucine. Orthologues of CYP71E1 are not readily apparent in any of the sequenced plant genomes.

As suggested by the numerous defense toxins derived from the phenylpropanoid pathway and the presence of CYP73A subfamily members in all plants, *p*-hydroxylation of *t*-cinnamic acid to produce *p*-coumaric acid is an early P450-mediated conversion essential to all plants. In the Apiaceae, which includes *Pastinaca sativa* (parsnip), *Ammi majus* (Bishop's weed) and *Apium graveolens* (celery), as well as

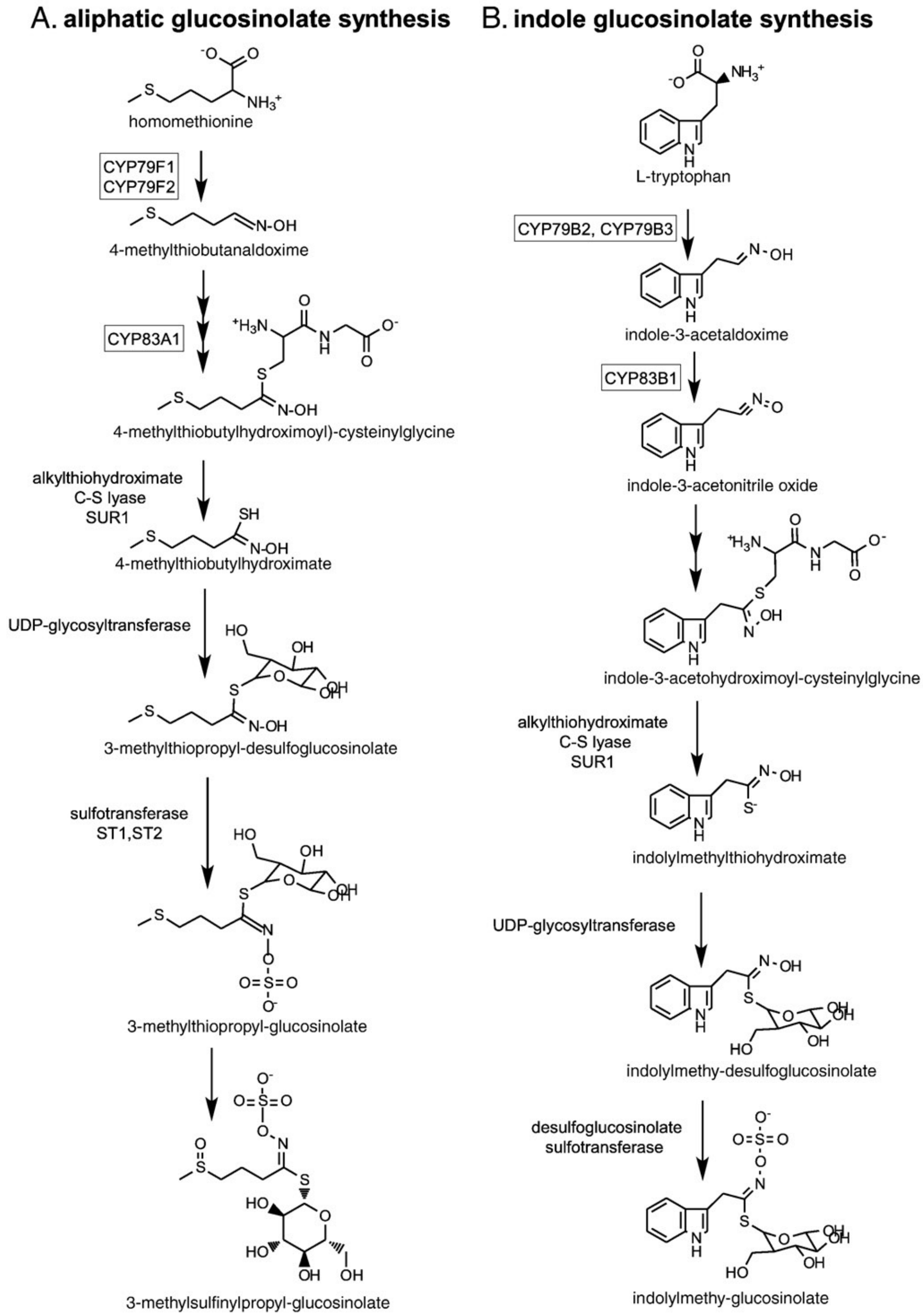

Fig. 2. Glucosinolate synthesis. The synthetic pathways for (A) aliphatic glucosinolates and (B) indole glucosinolates occurring in *Arabidopsis thaliana* are shown with boxes designating known P450 steps.

selected species of Rutaceae, Fabaceae and Moraceae, production of coumarins proceeds via the P450-mediated *o*-hydroxylation of *p*-coumaric acid to generate 2,4-dihydroxycinnamic acid and then to umbelliferone [59]. Subsequent modifications by prenyltranferases acting at alternate positions bifurcate the furanocoumarin pathway (Fig. 1A) to produce demethylsuberosin, the precursor for linear furanocoumarins, and osthenol, the precursor for angular furanocoumarins. Conversion of the 6-prenylated substrate to marmesin is mediated by an unidentified P450 and linear furanocoumarin synthesis proceeds using psoralen synthases (CYP71AJ1, CYP71AJ2, and CYP71AJ3) and O-methyltransferases [59–61]. Conversion of the 8-prenylated substrate to columbianetin is mediated by another unidentified P450 and angular furanocoumarin synthesis proceeds using a closely related angelicin synthase (CYP71AJ4). With obvious phylogenetic origins and relatively high overall amino acid identity (70–90%), these psoralen synthases and angelicin synthase mediate quite different catalytic reactions (formation of a dihydrofuran ring from a prenylated substrate vs. oxidative carbon–carbon chain cleavage) because of significant differences in four of their substrate recognition sites (SRS2, SRS3, SRS4, and SRS6) [61]. Once formed, psoralen and angelicin are further hydroxylated at several positions on the coumarin nucleus by other unidentified P450s yielding the wide array of complex linear and angular furanocoumarins (xanthotoxin, bergapten, and sphondin) present in the Apiaceae, Rutaceae and Fabaceae [59].

In *Sesame indicum* (sesame), the dimerization of the phenylpropanoid derivative pinoresinol to form the furofuran lignan sesamin containing two methylenedioxy bridges is mediated by CYP81Q1 [62]. Pinoresinol, which is itself formed by the coupling of conferyl alcohols, is also converted into sesamin by the related CYP81Q2 from *S. radiatum* (96% identical to CYP81Q1) but not by CYP81Q3 from *S. alatum*; divergences in the SRS2 and SRS3 regions of CYP81Q3 may account for these catalytic deficiencies. No other P450s involved in the synthesis of furofuran lignans have yet been cloned making it unclear whether they are members of this same subfamily. Those involved in the formation of methylenedioxy bridges in other groups of compounds, such as antimicrobial alkaloids, are categorized in an entirely different group of P450s. In *Coptis japonica* (Japanese goldthread), this includes CYP719A1 that mediates methylenedioxy bridge formation in berberine synthesis [63]. In *Eschscholzia californica* (California poppy), this includes several CYP719A subfamily members that mediate methylenedioxy bridge formation in stylopine synthesis and eventually lead to sanguinarine and macarpine synthesis [64,65]. Among these, CYP719A1 is 65% identical to CYP719A2 and CYP719A3 and just 24% amino acid identical to CYP81Q1 in furofuran lignan synthesis. P450s involved in other methylenedioxy bridge formations have not yet been identified.

In Coniferales (conifers), P450s involved in synthesizing and modifying complex mixtures of monoterpenes (myrcene, pinenes), sesquiterpenes (farnesene) and diterpenes (abietadienol and abietic acid) provide especially interesting perspectives on the chemical ecologies associated with plant–insect interactions. Conversion of geranylgeranyl diphosphate to C_{20} terpenoid skeletons, such as abietadiene and cadinene, by terpenoid synthases [66] is followed by multiple oxidation steps generating diterpenoid resin acids [12]. The multifunctional CYP720B1 and close relatives in this subfamily are capable of utilizing diterpenes, alcohols and aldehydes as substrates to produce large collections of diterpene olefins, alcohols, aldehydes and resin acids [67–69].

In *Zea mays* (maize), a series of four clustered P450 genes in the CYP71C subfamily synthesize benzoxazinoids such as DIMBOA that is broken down to toxic aglycons upon herbivore damage [70–72]. The fact that several other monocots and dicots synthesize benzoxazinoids indicates that this biosynthetic pathway has evolved independently several times [73] but the P450 complements mediating these other synthetic reactions have not yet cloned.

2.4. P450s in defense signaling pathways

Biosyntheses of these plant defense toxins are often dramatically activated by signaling pathways that also include P450s. Some of their most effective inducers, jasmonate (JA) and methyl jasmonate (MeJA), are cyclic oxidation products derived from linolenic acid via allene oxide synthases, which are unconventional P450s in the CYP74A subfamily [74,75]. Other inducers, including hexenal, related C_6-volatiles and octadecanoic acid, are cleavage products derived from the same polyunsaturated fatty acid via hydroperoxide lyases, which are in the CYP74B subfamily [76,77].

3. Insect P450s

3.1. Gene counts

Insects have clearly adapted to the presence of plant toxins in their diets. Among several enzyme groups capable of inactivating plant toxins, P450s are key mediators of the hydroxylations and epoxidations required for the efficient destruction and elimination of toxins prior to their adsorption in gut tissues. In insect genomes that have not undergone the level of gene duplication seen in plant genomes, the P450 gene superfamily includes 46 full-length genes in *Apis mellifera* (honey bee), 87 in *Drosophila melanogaster* (fruit fly), 83 in *D. pseudoobscura*, 105 in *Anopheles gambiae* (malarial mosquito), 153 in *Aedes aegypti* (dengue and yellow fever mosquito), 79 in *Bombyx mori* (silkworm), 91 in *Nasonia vitripennis* (parasitoid wasp) and 136 in *Tribolium casteneum* (red flour beetle) [41,78–82]. These and other sequences cloned more sporadically account for the current total of 1750 insect P450 sequences available in databases.

3.2. P450s in detoxicative pathways

Despite their critical importance in defining food sources for herbivores, the P450s responsible for the metabolism of plant allelochemicals have been studied in only a few insect species [5–7,83,84]. Those involved in the biotransformation of toxic plant furanocoumarins have been characterized in eight insect species (four *Papilio* species (swallowtails), *Helicoverpa zea* (corn earworm), *Depressaria pastinacella* (parsnip webworm), *Amyelois transitella* (navel orangeworm) and *A. gambiae*). In the specialist *P. polyxenes*, CYP6B1 metabolizes several, but not all, linear furanocoumarins efficiently (Fig. 1B) and angular furanocoumarins when supplemented with additional P450 reductase [85–89]. In the related generalists *P. glaucus* and *P. multicaudatus* that encounter wider ranges of toxins in their host plants, CYP6B4, CYP6B17 and CYP6B33 metabolize a broad range of linear and angular furanocoumarins [85,90–92].

In the generalist *H. zea*, multiple CYP6B subfamily members exist [93,94]. Of these, CYP6B8 metabolizes xanthotoxin (similar to *Papilio* CYP6B1, CYP6B4 and CYP6B33) as well as a variety of other plant compounds (flavone, α-naphthoflavone (α-NF), chlorogenic acid, indole-3-carbinol, quercetin, and rutin) and insecticides (cypermethrin (pyrethroid), diazinon (organophosphate), and aldrin (cyclodiene)) [95,96]. The predicted structures for the closely related CYP6B9 and CYP6B27 suggest that their substrate specificities are very similar to those determined for CYP6B8. In addition, *H. zea* also expresses CYP321A1, which metabolizes many of the same compounds as CYP6B8 (xanthotoxin, angelicin, α-NF, cypermethrin, diazinon, and aldrin) and an additional fungal toxin (aflatoxin B1) [96–98]. The existence of two different groups of P450s mediating similar metabolisms indicates that these allelochemical detoxification systems have evolved at least twice in these species.

In the highly specialized *D. pastinacella* that feeds solely on two furanocoumarin-containing plants and metabolizes furanocoumarins at very high rates (10-fold higher than any *Papilio* species) [99], CYP6AB3 selectively metabolizes one furanocoumarin (imperatorin) (Fig. 1C) and one natural MDP compound (myristicin), which are extremely abundant in its host plants [100–102]. With just these two demonstrated activities and no ability to metabolize other furanocoumarins, CYP6AB3 is probably the most specialized P450 that can catabolize plant toxins. In *Amyelois transitella* (navel orangeworm) that devastates almond, fig and pistachio nuts, a closely related CYP6AB11 also metabolizes significant amounts of imperatorin but not myristicin [103]. Reflecting the high diversity of P450s metabolizing furanocoumarins: *D. pastinacella* CYP6AB3 shares 55% amino acid identity with *A. transitella* CYP6AB11 and 36–38% identity with members of the CYP6B subfamily; *A. transitella* CYP6AB11 shares 35–38% identity with the CYP6B subfamily, 55% identity with *D. pastinacella* CYP6AB3 and 67% identity with *B. mori* CYP6AB4; *H. zea* CYP321A1 shares 30–32% amino acid identity with members of the CYP6B subfamily. Additional CYP6B subfamily members identified in *D. pastinacella* and *A. transitella* have not yet been demonstrated to metabolize furanocoumarins.

Other biochemically characterized insect P450s capable of metabolizing plant allelochemicals include *M. domestica* CYP6A1 that metabolizes plant terpenoids [104], *Diploptera punctata* (Pacific beetle cockroach) CYP4C7 that hydroxylates juvenile hormone III (JHIII) and other sesquiterpenoids [105], *A. mellifera* CYP6AS members that metabolize quercetin (flavonoid) [106], *A. gambiae* CYP6Z1 that metabolizes furanocoumarins (xanthotoxin and bergapten), furanochromones (khellin and visnagin) and natural MDP compounds (myristicin, safrole and isosafrole) [107,108], *A. gambiae* CYP6Z2 that metabolizes xanthotoxin, enterodiol (lignan), resveratol and piceatannol (stilbenes) [109].

In addition to these examples, P450s in various Coleoptera species including *Ips pini* (pine engraver beetles) and *Ips paraconfusus* (California fivespined ips) have evolved fairly novel strategies for dealing with the especially high concentrations of monoterpenes, sesquiterpenes and diterpenoid resin acids encountered in their feeding and reproductive cycles, which occur in conifer barks. In addition to the expected P450-mediated detoxification of the monoterpene myrcene (by an unidentified P450), male bark beetles ingesting this monoterpene can use it as substrate for production of aggregation pheromone, a mixture of ipsdienol, ipsenol and other compounds that recruits other beetles to damaged trees and enhances the reproductive success of individual *Ips* species [110]. Not entirely willing to rely on exposure to plant-derived myrcene, male bark beetles synthesize myrcene *de novo* and then convert it into ipsdienol and ipsenol [110]. Molecular analysis of the *I. paraconfusus* P450 transcripts induced in response to pine bark mixtures identified several inducible CYP4 family members and one CYP9T subfamily member that are differentially expressed in males and females or else constitutively expressed in both sexes [111]. Similar analysis of *I. pini* P450 transcripts identified a closely related CYP9T subfamily member that is differentially expressed in males [112]. Biochemical characterizations of the *I. pini* CYP9T2 and *I. paraconfusus* CYP9T1 proteins subsequently indicated that, while both mediate production of ipsdienol racemers, the final proportions of these in various aggregation pheromone components are dictated by later enzymes in their metabolic pathways and not by either of these P450s [112,113]. Similar transcript analysis in *Dendroctonus poderosae* (mountain pine beetle) has now identified CYP6CR1 as male-specific and inducible and implicated it in the production of this insect's male aggregation pheromone component, *exo*-brevicomin [114]. Although not yet biochemically characterized, it is suggested that CYP6CR1 mediates the epoxidation of a fatty acid precursor to 6,7-epoxynonan-2-one and that another unidentified P450 mediates hydroxylation of ingested α-pinene to verbenol.

In several cases, less direct evidence has implicated particular P450s in the metabolism of plant allelochemicals. In one recent example, transgenic plant-mediated RNAi silencing of *H. armigera* CYP6AE14 transcripts reduced larval tolerance to the sesquiterpene gossypol [115]. In other examples, inductions of CYP4D and CYP28A subfamily transcripts in cactophilic *Drosophila* species (*D. metterli*, *D. mojavensis*, *D. pachea* and *D. nigrospiracula*) exposed to toxic isoquinoline alkaloids have suggested that these particular P450s catabolize these compounds [116–118]. While differentially expressed by the isoquinoline alkaloids in their cactus hosts [117,118], phylogenetic analyses of the CYP28A1 and CYP4D10 genes have indicated that only the CYP28A1 sequence has adaptively evolved [119]. In the last example, inductions of CYP6A8, CYP6D5, CYP6W1, CYP9B2 and CYP12D1 transcripts in *D. melanogaster* exposed to *Piper nigrum* (black pepper) extracts have suggested that one or more of these are involved in catabolism of piperamides [120]. It is also possible that some of these proteins metabolize the assortment of MDP compounds present along with piperamides in *Piper* species.

3.3. Induction of detoxicative P450s

As noted previously, ingestion of plant toxins frequently causes insects to induce P450 transcripts responsible for catabolism of plant toxins, a feed-forward loop important for optimal survival as toxins are encountered in new plant groups. In the specialist *P. polyxenes*, *CYP6B1* transcripts, which code for furanocoumarin metabolism, are induced by the linear furanocoumarin xanthotoxin and to a lesser extent by the angular furanocoumarin angelicin [121–123]. Related *CYP6B3* transcripts, which also code for furanocoumarin metabolism, are induced by a wide range of linear and angular furanocoumarins [123,124] providing these species with a broad inducible response to these group of toxins. In the generalist *H. zea*, *CYP6B8* transcripts, which code for detoxification of a broad range of substrates, are strongly induced by xanthotoxin and imperatorin (furanocoumarins), indole-3-carbinol, chlorogenic acid (shikimate intermediate), flavone (flavone aglycone), jasmonic acid and salicylic acid (plant signaling molecules) and more weakly by gossypol (sesquiterpene), rutin (flavonol glycoside), visnagin (furanochromone), coumarin and quercetin (flavonol aglycone) [125–128]. *CYP321A1* transcripts, which also code for detoxification of a broad range of substrates, are also induced by many of these same allelochemicals [97,127,128]. In the closely related *H. armigera*, *CYP9A* subfamily transcripts are induced by gossypol, deltamethrin (pyrethroid) and phenobarbital (PB, barbiturate drug); other plant xenobiotics have not been tested [129]. Consistent with their suggested involvement in gossypol metabolism, *H. armigera CYP6AE14* transcripts are selectively induced by gossypol but not any terpenes or phenolics tested [115]. In the Solanaceae specialist *Manduca sexta* (tobacco hornworm), *CYP6M* and *CYP9A* subfamily transcripts are induced by 2-undecanone and 2-tridecanone (aliphatic ketones), indole 3-carbinol and xanthotoxin [130,131]. In *D. melanogaster*, *CYP6A2*, *CYP6A8* and multiple other P450 transcripts are induced by caffeine (xanthine alkaloid), PB and PBO [132–134]. In this same species, *CYP6A8*, *CYP12D1* and several other P450 transcripts are induced by black pepper extracts [120]. In *A. aegypti*, multiple *CYP4*, *CYP6* and *CYP9* family transcripts are induced in response to the allelochemical combinations in leaf litter [135]. In *A. mellifera*, several *CYP6AS* subfamily transcripts are induced by honey extracts [136]. Other examples of plant allelochemicals inducing insect P450 transcripts are summarized in Feyereisen [5].

In addition to these previous examples, several studies focusing solely on environmental xenobiotics have identified a few other synthetic compounds capable of inducing insect P450 transcripts. In *A. aegypti*, fluoranthene (arylhydrocarbon), permethrin and temephos (insecticides) and atrazine (herbicide) induce various *CYP6* and *CYP9* subfamily transcripts [137,138]. In *D. melanogaster*, dichlorodiphenyltrichloroethane (DDT) induces *CYP12D1* transcripts [139]. In *M. domestica*, permethrin induces several *CYP4* and *CYP6* family transcripts [140].

3.4. Inhibition of detoxicative P450s

Given that insects are exposed to varying mixtures of these environmental xenobiotics and host plant allelochemicals, P450 transcriptional responses are inevitably complex. This complexity is further complicated by the fact that some compounds in these mixtures induce expression of detoxicative P450 transcripts while other compounds inhibit their enzymatic functions. These inhibitory effects are clearly evident in the prominent use of PBO, the synthetic MDP compound, as an insecticide synergist intended to block P450-mediated metabolism of insecticides [141]. They are also evident in the use of natural mixtures of plant compounds, such as the piperamides from *Piper* species, as synergists [142].

Detailed analyses of the range of natural plant compounds interfering with metabolism have been conducted in only a few studies with heterologously expressed insect P450 proteins. Among these, furanocoumarin metabolism by *P. polyxenes* CYP6B1 is inhibited by a range of compounds including coumarins, MDPs (myristicin, safrole, and isosafrole), furanochromones (visnagin and khellin), flavonoids (flavone and α-napthoflavone), alkaloids (pilocarpine) [143,144]. Comparisons among these inhibition profiles have indicated that the three natural MDP compounds inhibit CYP6B1 more efficiently than synthetic PBO and provided the first concrete evidence that PBO does not block all insect

P450 activities [144]. Benzyloxyresorufin (model substrate) metabolism by *A. gambiae* CYP6Z2 is inhibited by xanthotoxin, α-napthoflavone, quercitin (flavonol), resveratol (stilbene), genistein (isoflavone) and cypermethrin (pyrethroid) [109].

3.5. Catalytic site evolution

While comparatively few insect P450s metabolizing plant allelochemicals have been studied, catalytic sites in some of these have been analyzed in enough detail to understand the evolutionary adaptations that have allowed some to acquire new metabolic capabilities for toxins. Comparisons between the catalytic site in *P. polyxenes* CYP6B1, which metabolizes a select number of furanocoumarins, and *H. zea* CYP6B8, which metabolizes a broad range of plant allelochemicals and insecticides, have indicated that multiple aromatic residues in SRS2, SRS5 and SRS6 of CYP6B1 are replaced by nonaromatic residues in CYP6B8 [95,96]. These and additional polar-to-nonpolar replacements in the hydrophilic cage surrounding the aromatic network responsible for substrate positioning in CYP6B1 [143] are apparently sufficient to enhance catalytic site flexibility in CYP6B8 and account for its metabolism of a wide range of plant compounds and insecticides [95,96]. Similarly, comparisons of the catalytic site in *P. polyxenes* CYP6B1 with those in *P. glaucus* CYP6B4 and *P. canadensis* CYP6B33, which metabolize broader ranges of substrates, have shown that replacements of large catalytic site residues with smaller ones increases catalytic site volumes enough to accommodate the wider range of allelochemicals ingested by these polyphagous species [90,91]. Comparisons of the catalytic sites in *A. gambiae* CYP6Z1, which metabolizes a range of allelochemical and insecticide classes, and CYP6Z2, which metabolizes only a few compounds, have indicated that as few as one or two catalytic site replacements in SRS2 and SRS4 account for the broader substrate range of CYP6Z1 [107].

4. Conclusions

In summary, synthesis of the large array of defense toxins and other secondary metabolites in plants is accomplished by an astounding number of P450 proteins that have been derived from continual duplication and divergence of ancestral genes. As further biochemical characterizations of these proteins places them in particular biosynthetic pathways, the many types of molecular events associated with the evolution of plant pathways for the production of new defense toxins should become more apparent than currently. Our existing knowledge of plant P450s has already indicated that: (1) many P450 families and subfamilies are being utilized for production of defense toxins, (2) divergence in catalytic site residues of duplicated P450 gene sets can lead to novel defense toxins and (3) selective transcription limits production of defense toxins to tissues and conditions in which they are most needed. Countering this trend of increasing chemical diversity in plant evolution, biochemical characterizations of insect P450s capable of allelochemical metabolism have already indicated that: (1) relatively few P450 families and subfamilies used for the elimination of defense toxins, (2) divergence in catalytic site residues in existing P450 genes can substantially broaden substrate range and (3) selective transcription limits production of detoxicative enzymes to conditions in which they are most needed. Future analyses of P450s involved in plant–insect interactions will undoubtedly expand on the few P450 examples cited in this review and begin to describe novel biosynthetic and detoxicative functions not yet characterized. Coupled with adequate information on the range of inducers and inhibitors for each, these studies will provide critical information needed for further genetic manipulation of plant species for the production of transgenic plants with enhanced resistance to insects and genetic manipulation of insect species for inactivation of the monooxygenase activities mediating allelochemical and insecticide detoxifications. Recognizing that the combined collection of plant and insect P450s far outnumbers those in other organisms and that many molecular, biochemical and computational tools are available for their studies, the future is bright for students and faculty versed in the technologies needed to define and manipulate P450s.

Acknowledgements

The author acknowledges her long-term collaboration with Dr. May Berenbaum that has been integral to our detailing of biochemical functions in insect P450s. She also acknowledges the contributions of Dr. Sanjeewa Rupasinghe that have provided molecular perspectives on the catalytic sites of innumerable plant and insect P450s and the contributions of Ms. Iyoncy Rodrigo in many areas of P450 genomics and manuscript development. She also thanks the very long list of graduate students, postdoctoral fellows and research associates that have contributed to functional and transcriptional P450 studies in her laboratory and the funding of these by the National Institutes of Health, the National Science Foundation and the United States Department of Agriculture Competitive Grants Program.

References

[1] D. Werck-Reichhart, S. Bak, S. Paquette, Cytochrome P450, in: C.R. Somerville, E.M. Meyerowitz (Eds.), The Arabidopsis Book, American Society of Plant Biologists, Rockville, MD, 2002, doi:10.1199/tab.0028, http://www.aspb.org/publications/arabidopsis.
[2] K.A. Nielsen, B.L. Moller, Cytochrome P450s in Plants, in: P.R.O. de Montellano (Ed.), Cytochrome P450: structure, mechanism, and biochemistry, 3rd ed., Kluwer Academic/Plenum Publishers, New York, 2005, pp. 553–583.
[3] M.A. Schuler, H. Duan, M. Bilgin, S. Ali, *Arabidopsis* P450s through the looking glass: a window on plant biochemistry, Phytochem. Rev. 5 (2006) 205–237.
[4] M. Mizutani, D. Ohta, Diversification of P450 genes during land plant evolution, Ann. Rev. Plant Biol. 61 (2010) 25.1–25.25.
[5] R. Feyereisen, in: L.I. Gilbert, K. Latrou, S.S. Gill (Eds.), Comprehensive Molecular Insect Science, vol. 4, Elsevier, Oxford, 2005, pp. 1–77.
[6] X. Li, M.A. Schuler, M.R. Berenbaum, Molecular mechanisms of metabolic resistance to synthetic and natural xenobiotics, Ann. Rev. Entomol. 52 (2007) 231–253.
[7] L. Despres, J.P. David, C. Gallet, The evolutionary ecology of insect resistance to plant chemicals, Trends Ecol. Evol. 22 (2007) 298–307.
[8] D.S. Seigler, Plant Secondary Metabolism, Kluwer Academic Publishers, Boston, MA, 1998, p. 776.
[9] M. Wink (Ed.), Functions and Biotechnology of Plant Secondary Metabolites, 2nd ed., Annual Plant Reviews, vol. 39, Blackwell Publishing Ltd., 2010, pp. 1–20.
[10] S. Wu, J. Chappell, Metabolic engineering of natural products in plants; tools of the trade and challenges for the future, Curr. Opin. Biotechnol. 19 (2008) 145–152.
[11] D.J. McGarvey, R. Croteau, Terpenoid metabolism, Plant Cell 7 (1995) 1015–1026.
[12] K.G. Zulak, J. Bohlmann, Terpenoid biosynthesis and specialized vascular cells of conifer defense, J. Integr. Plant Biol. 52 (2010) 86–97.
[13] T.M. Kutchan, Alkaloid biosynthesis-the basis for metabolic engineering of medicinal plants, Plant Cell 7 (1995) 1059–1070.
[14] P.J. Facchini, Alkaloid biosynthesis in plants: biochemistry, cell biology, molecular regulation, and metabolic engineering applications, Ann. Rev. Plant Physiol. Plant Mol. Biol. 52 (2001) 29–66.
[15] J. Ziegler, P.J. Facchini, Alkaloid biosynthesis: metabolism and trafficking, Ann. Rev. Plant Biol. 59 (2008) 735–769.
[16] R.D.H. Murray, J. Mendez, S.A. Brown, The natural coumarins, J. Wiley and Sons, Chicester, United Kingdom, 1982, p. 702.
[17] M.R. Berenbaum, Coumarins, in: G. Rosenthal, M. Berenbaum (Eds.), Herbivores: their interactions with secondary plant metabolites, vol. 1, Academic Press, San Diego, 1991, pp. 221–249.
[18] M. Curini, G. Cravotto, F. Epifano, G. Giannone, Chemistry and biological activity of natural and synthetic prenyloxycoumarins, Curr. Med. Chem. 13 (2006) 199–222.
[19] B.F. Abeysekera, Z. Abramowski, G.H. Towers, Genotoxicity of the natural furochromones, khellin and visnagin and the identification of a khellin-thymine photoadduct, Photochem. Photobiol. 38 (1983) 311–315.
[20] B.A. Halkier, J. Gershenzon, Biology and biochemistry of glucosinolates, Annu. Rev. Plant Biol. 57 (2006) 303–333.
[21] M. Nafisi, I.E. Sønderby, B.G. Hansen, F. Geu-Flores, H.H. Nour-Eldin, M.H.H. Nørholm, N.B. Jensen, J. Li, B.A. Halkier, Cytochromes P450 in the biosynthesis of glucosinolates and indole alkaloids, Phytochem. Rev. 5 (2006) 331–346.
[22] I.M. Scott, H.R. Jensen, B.J.R. Philogène, J.T. Arnason, A review of Piper spp. (*Piperaceae*) phytochemistry, insecticidal activity and mode of action, Phytochem. Rev. 7 (2008) 65–75.
[23] A. Gierl, M. Frey, Evolution of benzoxazinone biosynthesis and indole production in maize, Planta 213 (2001) 493–498.

[24] M. Zagrobelny, S. Bak, A.V. Rasmussen, B. Jørgensen, C.M. Naumann, B.L. Møller, Cyanogenic glucosides and plant–insect interactions, Phytochemistry 65 (2004) 293–306.

[25] S. Bak, S.M. Paquette, M. Morant, A.V. Morant, S. Saito, N. Bjarnholt, M. Zagrobelny, K. Jørgensen, S. Osmani, H.T. Simonsen, R.S. Pérez, T.B. van Heeswijck, B. Jørgensen, B.L. Møller, Cyanogenic glycosides: a case study for evolution and application of cytochromes P450, Phytochem. Rev. 5 (2006) 309–329.

[26] M. Murray, Mechanisms of inhibitory and regulatory effects of methylenedioxyphenyl compounds on cytochrome P450-dependent drug oxidation, Curr. Drug Metab. 1 (2000) 67–84.

[27] T. Umezawa, Diversity in lignan biosynthesis, Phytochem. Rev. 2 (2003) 371–390.

[28] P. Hodek, P. Trefil, M. Stiborova, Flavonoids-potent and versatile biologically active compounds interacting with cytochromes P450, Chem. Biol. Interact. 139 (2002) 1–21.

[29] Y.J. Moon, X. Wang, M.E. Morris, Dietary flavonoids: effects on xenobiotic and carcinogen metabolism, Toxicol. In Vitro 20 (2006) 187–210.

[30] B.S.J. Winkel, The Biosynthesis of Flavonoids, in: E. Grotewold (Ed.), The Science of flavonoids, Springer, 2006, pp. 71–95.

[31] F. Ververidis, E. Trantas, C. Douglas, G. Vollmer, G. Kretzschmar, N. Panopoulos, Biotechnology of flavonoids and other phenylpropanoid-derived natural products. Part I: chemical diversity, impacts on plant biology and human health, Biotechnol. J. 2 (2007) 1214–1234.

[32] R.A. Dixon, N. Paiva, Stress-induced phenylpropanoid metabolism, Plant Cell 7 (1995) 1085–1097.

[33] P. Jeandet, A.-C. Douillet-Breuil, R. Bessis, S. Debord, M. Sbaghi, M. Adrian, Phytoalexins from the Vitaceae: biosynthesis, phytoalexin gene expression in transgenic plants, antifungal activity, and metabolism, J. Agric. Food Chem. 50 (2002) 2731–2741.

[34] G.J. Wood, B. Rogina, S. Lavu, K. Howitz, S.L. Helfand, M. Tatar, D. Sinclair, Sirtuin activators mimic caloric restriction and delay ageing in metazoans, Nature 430 (2004) 686–689.

[35] M.R. Berenbaum, A.R. Zangerl, Phytochemical diversity: Adaptation or random variation? In: phytochemical diversity and redundancy in ecological interactions, in: J. Romeo, et al., (Eds.), Rec. Adv. Phytochem., 30, Plenum Press, New York, 1996, pp. 1–24.

[36] S.M. Paquette, S. Bak, R. Feyereisen, Intron-exon organization and phylogeny in a large superfamily, the paralogous cytochrome P450 genes of *Arabidopsis thaliana*, DNA Cell Biol. 19 (2000) 307–317.

[37] S.M. Paquette, K. Jensen, S. Bak, A web-based resource for the Arabidopsis P450, cytochromes b5, NADPH-cytochrome P450 reductases, and family 1 glycosyltransferases (http://www.P450.kvl.dk), Phytochemistry 70 (2009) 1940–1947.

[38] D.R. Nelson, M.A. Schuler, S.M. Paquette, D. Werck-Reichhart, S. Bak, Comparative genomics of *Oryza sativa* and *Arabidopsis thaliana*. Analysis of 727 chromosome P450 genes and pseudogenes from a monocot and a dicot, Plant Physiol. 135 (2004) 756–772.

[39] D.R. Nelson, R. Ming, M. Alam, M.A. Schuler, Comparison of cytochrome P450 genes from six plant genomes, Trop. Plant Biol. 1 (2008) 216–235.

[40] D.R. Nelson, Plant cytochrome P450s from moss to poplar, Phytochem. Rev. 5 (2006) 193–204.

[41] D.R. Nelson, The cytochrome P450 homepage, Hum. Genomics 4 (2009) 59–65.

[42] C.H. Hansen, U. Wittstock, C.E. Olsen, A.J. Hick, J.A. Pickett, B.A. Halkier, Cytochrome P450 CYP79F1 from Arabidopsis catalyzes the conversion of dihomomethionine and trihomomethionine to the corresponding aldoximes in the biosynthesis of aliphatic glucosinolates, J. Biol. Chem. 276 (2001) 11078–11085.

[43] B. Reintanz, M. Lehnen, M. Reichelt, J. Gershenzon, M. Kowalczyk, G. Sandberg, M. Godde, R. Uhl, K. Palme, *bus*, a bushy Arabidopsis *CYP79F1* knockout mutant with abolished synthesis of short-chain aliphatic glucosinolates, Plant Cell 13 (2001) 351–367.

[44] S.-X. Chen, E. Glawischnig, K. Jørgensen, P. Naur, B. Jørgensen, C.E. Olsen, C.H. Hansen, H. Rasmussen, J.A. Pickett, B.A. Halkier, Cytochrome P450 CYP79F1 and CYP79F2 genes catalyze the first step in the biosynthesis of short-chain and long-chain aliphatic glucosinolates in Arabidopsis, Plant J. 33 (2003) 923–937.

[45] S. Bak, R. Feyereisen, The involvement of two P450 enzymes, CYP83B1 and CYP83A1, in auxin homeostasis and glucosinolate biosynthesis, Plant Physiol. 127 (2001) 108–118.

[46] M.R. Hemm, M.O. Ruegger, C. Chapple, The Arabidopsis *ref2* mutant is defective in the gene encoding CYP83A1 and shows both phenylpropanoid and glucosinolate phenotypes, Plant Cell 15 (2003) 179–194.

[47] P. Naur, C.H. Hansen, S. Bak, B.G. Hansen, N.B. Jensen, H.L. Nielsen, B.A. Halkier, CYP79B1 from *Sinapis alba* converts tryptophan to indole-3-acetaldoxime, Arch. Biochem. Biophys. 409 (2003) 235–241.

[48] A.K. Hull, R. Vij, J.L. Celenza, Arabidopsis cytochrome P450s that catalyze the first step of tryptophan-dependent indole-3-acetic acid biosynthesis, Proc. Natl Acad. Sci. USA 97 (2000) 2379–2384.

[49] M.D. Mikkelsen, C.H. Hansen, U. Wittstock, B.A. Halkier, Cytochrome P450 CYP79B2 from Arabidopsis catalyzes the conversion of tryptophan to indole-3-acetaldoxime, a precursor of indole glucosinolates and indole-3-acetic acid, J. Biol. Chem. 275 (2000) 33712–33717.

[50] S. Bak, F.E. Tax, K.A. Feldmann, D.W. Galbraith, R. Feyereisen, CYP83B1, a cytochrome P450 at the metabolic branch point in auxin and indole glucosinolate biosynthesis in Arabidopsis, Plant Cell 13 (2001) 101–111.

[51] U. Wittstock, B.A. Halkier, Cytochrome P450 CYP79A2 from *Arabidopsis thaliana* L. catalyzes the conversion of L-phenylalanine to phenylacetaldoxime in the biosynthesis of benzylglucosinolate, J. Biol. Chem. 275 (2000) 14659–14666.

[52] B.A. Halkier, H.L. Nielsen, B. Koch, B.L. Møller, Purification and characterization of recombinant cytochrome P450TYR expressed at high levels in *Escherichia coli*, Arch. Biochem. Biophys. 322 (1995) 369–377.

[53] B.M. Koch, O. Sibbesen, B.A. Halkier, I. Svendsen, B.L. Møller, The primary sequence of cytochrome P450tyr, the multifunctionalN-hydroxylase catalyzing the conversion of L-tyrosine top-hydroxyphenylacetaldehyde oxime in the biosynthesis of the cyanogenic glucoside dhurrin in *Sorghum bicolor* (L.) Moench, Arch. Biochem. Biophys. 323 (1995) 177–186.

[54] S. Bak, R.A. Kahn, H.L. Nielsen, B.L. Møller, B.A. Halkier, Cloning of three A-type cytochromes P450, CYP71E1, CYP98, and CYP99 from *Sorghum bicolor* (L.) Moench by a PCR approach and identification by expression in *Escherichia coli* of CYP71E1 as a multifunctional cytochrome P450 in the biosynthesis of the cyanogenic glucoside dhurrin, Plant Mol. Biol. 36 (1998) 393–405.

[55] S. Bak, H.L. Nielsen, B.A. Halkier, The presence of CYP79 homologues in glucosinolate-producing plants shows evolutionary conservation of the enzymes in the conversion of amino acid to aldoxime in the biosynthesis of cyanogenic glucosides and glucosinolates, Plant Mol. Biol. 38 (1998) 725–734.

[56] M.D. Andersen, P.K. Busk, I. Svendsen, B.L. Møller, Cytochromes P-450 from cassava (*Manihot esculenta* Crantz) catalyzing the first steps in the biosynthesis of the cyanogenic glucosides linamarin and lotaustralin, J. Biol. Chem. 275 (2000) 1966–1975.

[57] J.S. Nielsen, B.L. Møller, Cloning and expression of cytochrome P450 enzymes catalyzing the conversion of tyrosine to *p*-hydroxyphenylacetaldoxime in the biosynthesis of cyanogenic glucosides in *Triglochin maritime*, Plant Physiol. 122 (2000) 1311–1321.

[58] K. Forslund, M. Morant, B. Jørgensen, C.E. Olsen, E. Asamizu, S. Sato, S. Tabata, S. Bak, Biosynthesis of the mitrile glucosides rhodiocyanoside A and D and the cyanogenic glucosides lotaustralin and linamarin in *Lotus japonicus*, Plant Physiol. 135 (2004) 71–84.

[59] F. Bourgaud, A. Hehn, R. Larbat, S. Doerper, E. Gontier, S. Kellner, U. Matern, Biosynthesis of coumarins in plants: a major pathway still to be unravelled for cytochrome P450 enzymes, Phytochem. Rev. 5 (2006) 293–308.

[60] R. Larbat, S. Kellner, S. Specker, A. Hehn, E. Gontier, J. Hans, F. Bourgaud, U. Matern, Molecular cloning and functional characterization of psoralen synthase, the first committed monooxygenase of furanocoumarin biosynthesis, J. Biol. Chem. 282 (2007) 542–554.

[61] R. Larbat, A. Hehn, J. Hans, S. Schneider, H. Jugdé, B. Schneider, U. Matern, F. Bourgaud, Isolation and functional characterization of CYP71AJ4 encoding for the first P450 monooxygenase of angular furanocoumarin biosynthesis, J. Biol. Chem. 284 (2009) 4776–4785.

[62] E. Ono, M. Nakai, Y. Fukui, N. Tomimori, M. Fukuchi-Mizutani, M. Saito, H. Satake, T. Tanaka, M. Katsuta, T. Umezawa, Y. Tanaka, Formation of two methylenedioxy bridges by a Sesamum CYP81Q protein yielding a furofuran lignan, (+)-sesamin, Proc. Natl Acad. Sci. USA 103 (2006) 10116–10121.

[63] N. Ikezawa, M. Tanaka, M. Nagayoshi, R. Shinkyo, T. Sakaki, K. Inouye, F. Sato, Molecular cloning and characterization of CYP719, a methylenedioxy bridge-forming enzyme that belongs to a novel P450 family, from cultured *Coptis japonica* cells, J. Biol. Chem. 278 (2003) 38557–38565.

[64] N. Ikezawa, K. Iwasa, F. Sato, Molecular cloning and characterization of methylenedioxy bridge-forming enzymes involved in stylopine biosynthesis in *Eschscholzia californica*, Febs J. 274 (2007) 1019–1035.

[65] N. Ikezawa, K. Iwasa, F. Sato, CYP719A subfamily of cytochrome P450 oxygenases and isoquinoline alkaloid biosynthesis in *Eschscholzia californica*, Plant Cell Rep. 28 (2009) 123–133.

[66] J. Bohlmann, G. Meyer-Gauen, R. Croteau, Plant terpenoid synthases: molecular biology and phylogenetic analysis, Proc. Natl Acad. Sci. USA 95 (1998) 4126–4133.

[67] D.K. Ro, G. Arimura, S.Y. Lau, E. Piers, J. Bohlmann, Loblolly pine abietadienol/abietadienal oxidase PtAO (CYP720B1) is a multifunctional, multisubstrate cytochrome P450 monooxygenase, Proc. Natl Acad. Sci. USA 102 (2005) 8060–8065.

[68] B. Hamberger, J. Bohlmann, Cytochrome P450 mono-oxygenases in conifer genomes: discovery of members of the terpenoid oxygenase superfamily in spruce and pine, Biochem. Soc. Trans. 34 (2006) 1209–1214.

[69] B. Hamberger, D. Hall, M. Yuen, C. Oddy, B. Hamberger, C.I. Keeling, C. Ritland, K. Ritland, J. Bohlmann, Targeted isolation, sequence assembly and characterization of two white spruce (*Picea glauca*) BAC clones for terpenoid synthase and cytochrome P450 genes involved in conifer defence reveal insights into a conifer genome, BMC Plant Biol. 9 (2009) 106.

[70] M. Frey, P. Chomet, E. Glawischnig, C. Stettner, S. Grün, A. Winklmair, W. Eisenreich, A. Bacher, R.B. Meeley, S.P. Briggs, K. Simcox, A. Gierl, Analysis of a chemical plant defense mechanism in grasses, Science 277 (1997) 696–699.

[71] M. Frey, K. Schullehner, R. Dick, A. Fiesselmann, A. Gierl, Benzoxazinoid biosynthesis, a model for evolution of secondary metabolic pathways in plants, Phytochemistry 70 (2009) 1645–1651.

[72] E. Glawischnig, S. Grün, M. Frey, A. Gierl, Cytochrome P450 monooxygenases of DIBOA biosynthesis: specificity and conservation among grasses, Phytochemistry 50 (1999) 925–930.

[73] K. Schullehner, R. Dick, F. Vitzthum, W. Schwab, W. Brandt, M. Frey, A. Gierl, Benzoxazinoid biosynthesis in dicot plants, Phytochemistry 69 (2008) 2668–2677.

[74] R.A. Creelman, J.E. Mullet, Oligosaccharins, brassinolides, and jasmonates: nontraditional regulators of plant growth, development and gene expression, Plant Cell 9 (1997) 1211–1223.

[75] A.R. Brash, Mechanistic aspects of CYP74 allene oxide synthases and related cytochrome P450 enzymes, Phytochemistry 70 (2009) 1522–1531.
[76] M.A. Noordermeer, G.A. Veldink, J.F.G. Vliegenthart, Fatty acid hydroperoxide lyase: a plant cytochrome P450 enzyme involved in wound healing and pest resistance, Chembiochem 2 (2001) 494–504.
[77] K. Matsui, Green leaf volatiles: hydroperoxide lyase pathway of oxylipin metabolism, Curr. Opin. Plant Biol. 9 (2006) 274–280.
[78] N. Tijet, C. Helvig, R. Feyereisen, The cytochrome P450 gene superfamily in *Drosophila melanogaster*: annotation, intron-exon organization and phylogeny, Gene 262 (2001) 189–198.
[79] C. Claudianos, H. Ranson, R.M. Johnson, S. Biswas, M.A. Schuler, M.R. Berenbaum, R. Feyereisen, J.G. Oakeshott, A deficit of detoxification enzymes: pesticide sensitivity and environmental response in the honeybee, Insect Mol. Biol. 15 (2006) 615–636.
[80] Tribolium Genome Sequencing Consortium, The genome of the model beetle and pest *Tribolium castaneum*, Nature 452 (2008) 949–955.
[81] C. Strode, C.S. Wondji, J.P. David, N.J. Hawkes, N. Lumjuan, D.R. Nelson, D.R. Drane, S.H. Karunaratne, J. Hemingway, W.C. Black, H. Ranson, Genomic analysis of detoxification genes in the mosquito *Aedes aegypti*, Insect Biochem. Mol. Biol. 38 (2008) 113–123.
[82] J.G. Oakeshott, R.M. Johnson, M.R. Berenbaum, H. Ranson, A.S. Cristino, C. Claudianos, Metabolic enzymes associated with xenobiotic and chemosensory responses in *Nasonia vitripennis*, Insect Mol. Biol. 1 (2010) 147–163.
[83] R. Feyereisen, Insect P450 enzymes, Annu. Rev. Entomol. 44 (1999) 507–533.
[84] T.G. Wilson, Resistance of Drosophila to toxins, Annu. Rev. Entomol. 46 (2001) 545–571.
[85] C.-F. Hung, M.R. Berenbaum, M.A. Schuler, Isolation and characterization of CYP6B4, furanocoumarin-inducible cytochrome P450 from a polyphagous caterpillar (Lepidoptera: *Papilionidae*), Insect Biochem. Mol. Biol. 27 (1997) 377–385.
[86] J.-S. Chen, M.R. Berenbaum, M.A. Schuler, Amino acids in SRS1 and SRS6 are critical for furanocoumarin metabolism by CYP6B1v1, a cytochrome P450 monooxygenase, Insect Mol. Biol. 11 (2002) 175–186.
[87] L. Pan, Z. Wen, M.R. Berenbaum, M.A. Schuler, Identification of variable amino acids in the SRS1 region of CYP6B1 modulating furanocoumarin metabolism, Arch. Biochem. Biophys. 22 (2004) 31–41.
[88] Z. Wen, L. Pan, M.R. Berenbaum, M.A. Schuler, Metabolism of linear and angular furanocoumarins by *Papilio polyxenes* CYP6B1 coexpressed with NADPH cytochrome P450 reductase, Insect Biochem. Mol. Biol. 33 (2003) 937–947.
[89] Z. Wen, S. Rupasinghe, G. Niu, M.R. Berenbaum, M.A. Schuler, CYP6B1 and CYP6B3 of the black swallowtail (*Papilio polyxenes*): adaptive evolution through subfunctionalization, Mol. Biol. Evol. 23 (2006) 2434–2443.
[90] W. Li, M.R. Berenbaum, M.A. Schuler, Diversification of furanocoumarin-metabolizing cytochrome P450s in two papilinoids: specificity and substrate encounter rate, Proc. Natl Acad. Sci. USA 100 (2003) 14593–14598.
[91] W. Mao, M.A. Schuler, M.R. Berenbaum, Cytochrome P450s in *Papilio multicaudatus* and the transition from oligophagy to polyphagy in the Papilionidae, Insect Mol. Biol. 16 (2007) 481–490.
[92] W. Mao, M.R. Berenbaum, M.A. Schuler, Modifications in the N-terminus of an insect P450 enhance production of catalytically active protein in baculovirus-Sf9 cell expression systems, Insect Biochem. Mol. Biol. 38 (2008) 66–75.
[93] X. Li, M.R. Berenbaum, M.A. Schuler, Molecular cloning and expression of CYP6B8: a xanthotoxin-inducible cytochrome P450 cDNA from *Helicoverpa zea*, Insect Biochem. Mol. Biol. 30 (2000) 75–84.
[94] X. Li, M.R. Berenbaum, M.A. Schuler, Cytochrome P450 and actin genes expressed in *Helicoverpa zea* and *Helicoverpa armigera*: paralogy/orthology identification, gene conversion, and evolution, Insect Biochem. Mol. Biol. 32 (2002) 311–320.
[95] X. Li, J. Baudry, M.R. Berenbaum, M.A. Schuler, Structural and functional evolution of insect CYP6B proteins: from specialist to generalist cytochrome P450, Proc. Natl Acad. Sci. USA 101 (2004) 2939–2944.
[96] S. Rupasinghe, Z. Wen, T.-L. Chiu, M.A. Schuler, *Helicoverpa zea* CYP6B8 and CYP321A1: different molecular solutions to the problem of metabolizing plant toxins and insecticides, Protein Eng. Des. Sel. 20 (2007) 615–624.
[97] M. Sasabe, Z. Wen, M.R. Berenbaum, M.A. Schuler, Molecular analysis of CYP321A1, a novel cytochrome P450 involved in metabolism of plant allelochemicals (furanocoumarins) and insecticides (cypermethrin) in *Helicoverpa zea*, Gene 338 (2004) 163–175.
[98] G. Niu, Z. Wen, S.G. Rupasinghe, R.S. Zeng, M.R. Berenbaum, M.A. Schuler, Aflatoxin B1 detoxification by CYP321A1 in *Helicoverpa zea*, Arch. Insect Biochem. Physiol. 69 (2008) 32–45.
[99] M.R. Berenbaum, Evolution of specialization in insect-umbellifer associations, Annu. Rev. Entomol. 35 (1990) 319–343.
[100] W. Li, A.R. Zangerl, M.A. Schuler, M.R. Berenbaum, Characterization and evolution of furanocoumarin-inducible cytochrome P450s in the parsnip webworm, *Depressaria pastinacella*, Insect Mol. Biol. 13 (2004) 603–613.
[101] W. Mao, S. Rupasinghe, A. Zangerl, M.A. Schuler, M.R. Berenbaum, Remarkable substrate-specificity of CYP6AB3 in *Depressaria pastinacella*, a highly specialized caterpillar, Insect Mol. Biol. 15 (2006) 169–179.
[102] W. Mao, A.R. Zangerl, M.R. Berenbaum, M.A. Schuler, Metabolism of myristicin by *Depressaria pastinacella* CYP6AB3v2 and inhibition by its metabolite, Insect Biochem. Mol. Biol. 38 (2008) 645–651.
[103] G. Niu, S.G. Rupasinghe, A.R. Zangerl, J.P. Siegel, M.A. Schuler, M.R. Berenbaum, A substrate-specific cytochrome P450 monooxygenase, CYP6AB11, from the polyphagous navel orangeworm (*Amyelois transitella*), Insect Biochem. Mol. Biol. in press.
[104] J.F. Andersen, J.K. Walding, P.H. Evans, W.S. Bowers, R. Feyereisen, Substrate specificity for the epoxidation of terpenoids and active site topology of house fly cytochrome P450 6A1, Chem. Res. Toxicol. 10 (1997) 156–164.
[105] T.D. Sutherland, G.C. Unnithan, J.F. Andersen, P.H. Evans, M.B. Murataliev, L.Z. Szabo, E.A. Mash, W.S. Bowers, R. Feyereisen, A cytochrome P450 terpenoid hydroxylase linked to the suppression of insect juvenile hormone synthesis, Proc. Natl Acad. Sci. USA 95 (1998) 12884–12889.
[106] W. Mao, S.G. Rupasinghe, R.M. Johnson, A.R. Zangerl, M.A. Schuler, M.R. Berenbaum, Quercetin-metabolizing CYP6AS enzymes of the pollinator *Apis mellifera* (Hymenoptera: Apidae), Comp. Biochem. Physiol. B Biochem. Mol. Biol. 154 (2009) 427–434.
[107] T.-L. Chiu, Z. Wen, S.G. Rupasinghe, M.A. Schuler, Comparative molecular modeling of *Anopheles gambiae* CYP6Z1, a mosquito P450 capable of metabolizing DDT, Proc. Natl Acad. Sci. USA 105 (2008) 8855–8860.
[108] S.G. Rupasinghe, Z. Wen, I.L. Rodrigo, M.R. Berenbaum, M.A. Schuler, CYP6Z1 from *Anopheles gambiae* metabolizes plant allelochemicals, insecticides and insecticide synergists: insights into mosquito controls by insecticides, in preparation.
[109] L.A. McLaughlin, U. Niazi, J. Bibby, J.P. David, J. Vontas, J. Hemingway, H. Ranson, M.J. Sutcliffe, M.J. Paine, Characterization of inhibitors and substrates of *Anopheles gambiae* CYP6Z2, Insect Mol. Biol. 17 (2008) 125–135.
[110] S.J. Seybold, D.P.W. Huber, J.C. Lee, A.D. Graves, J. Bohlmann, Pine monoterpenes and pine bark beetles: a marriage of convenience for defense and chemical communication, Phytochem. Rev. 5 (2006) 143–178.
[111] D.P.W. Huber, M.L. Erickson, C.M. Leutenegger, J. Bohlmann, S.J. Seybold, Isolation and extreme sex-specific expression of cytochrome P450 genes in the bark beetle, *Ips paraconfusus*, following feeding on the phloem of host ponderosa pine, *Pinus ponderosa*, Insect Mol. Biol. 16 (2007) 335–349.
[112] P. Sandstrom, W.H. Welch, G.J. Blomquist, C. Tittiger, Functional expression of a bark beetle cytochrome P450 that hydroxylates myrcene to ipsdienol, Insect Biochem. Mol. Biol. 36 (2006) 835–845.
[113] P. Sandstrom, M.D. Ginzel, J.C. Bearfield, W.H. Welch, G.J. Blomquist, C. Tittiger, Myrcene hydroxylases do not determine enantiomeric composition of pheromonal ipsdienol in Ips spp, J. Chem. Ecol. 34 (2008) 1584–1592.
[114] T. Aw, K. Schlauch, C.I. Keeling, S. Young, J.C. Bearfield, G.J. Blomquist, C. Tittiger, Functional genomics of mountain pine beetle (*Dendroctonus ponderosae*) midguts and fat bodies, BMC Genomics 11 (2010) 215.
[115] Y.-B. Mao, W.-J. Cai, J.-W. Wang, G.-J. Hong, X.-Y. Tao, L.-J. Wang, Y.-P. Huang, X.-Y. Chen, Silencing a cotton bollworm P450 monooxygenase gene by plant-mediated RNAi impairs larval tolerance of gossypol, Nat. Biotechnol. 25 (2007) 1307–1313.
[116] M.R. Frank, J.C. Fogleman, Involvement of cytochrome P450 in host-plant utilization by Sonoran Desert Drosophila, Proc. Natl Acad. Sci. USA 89 (1992) 11998–12002.
[117] P.B. Danielson, R.J. MacIntyre, J.C. Fogleman, Molecular cloning of a family of xenobiotic-inducible drosophilid cytochrome P450s: evidence for involvement in host-plant allelochemical resistance, Proc. Natl Acad. Sci. USA 94 (1997) 10797–10802.
[118] P.B. Danielson, J.L.M. Foster, M.M. McMahill, M.K. Smith, J.C. Fogleman, Induction by alkaloids and phenobarbital of Family 4 cytochrome P450s in *Drosophila*: evidence for involvement in host plant utilization, Mol. Gen. Genet. 259 (1998) 54–59.
[119] J.M. Bono, L.M. Matzkin, S. Castrezana, T.A. Markow, Molecular evolution and population genetics of two *Drosophila mettleri* cytochrome P450 genes involved in host plant utilization, Mol. Ecol. 17 (2008) 3211–3221.
[120] H.R. Jensen, I.M. Scott, S. Sims, V.L. Trudeau, J.T. Arnason, Gene expression profiles of *Drosophila melanogaster* exposed to an insecticidal extract of *Piper nigrum*, J. Agric. Food Chem. 54 (2006) 1289–1295.
[121] M.B. Cohen, M.A. Schuler, M.R. Berenbaum, A host-inducible cytochrome P450 from a host-specific caterpillar: molecular cloning and evolution, Proc. Natl Acad. Sci. USA 89 (1992) 10920–10924.
[122] C.-F. Hung, H. Prapaipong, M.R. Berenbaum, M.A. Schuler, Differential induction of cytochrome P450 transcripts in *Papilio polyxenes* by linear and angular furanocoumarins, Insect Biochem. Mol. Biol. 25 (1995) 89–99.
[123] R.A. Petersen, A.R. Zangerl, M.R. Berenbaum, M.A. Schuler, Expression of *CYP6B1* and *CYP6B3* cytochrome P450 monooxygenases and furanocoumarin metabolism in different tissues of *Papilio polyxenes*, Insect Biochem. Mol. Biol. 31 (2001) 679–690.
[124] C.-F. Hung, T.L. Harrison, M.R. Berenbaum, M.A. Schuler, CYP6B3: a second furanocoumarin-inducible cytochrome P450 expressed in *Papilio polyxenes*, Insect Mol. Biol. 4 (1995) 149–160.
[125] X. Li, M.R. Berenbaum, M.A. Schuler, Plant allelochemicals differentially regulate *Helicoverpa zea* cytochrome P450 genes, Insect Mol. Biol. 11 (2002) 343–351.
[126] X. Li, M.A. Schuler, M.R. Berenbaum, Jasmonate and salicylate induce transcription of herbivore cytochrome P450 genes, Nature 419 (2002) 712–715.
[127] Z. Wen, R.S. Zeng, G. Niu, M.R. Berenbaum, M.A. Schuler, Ecological significance of induction of broad-substrate P450s by natural and synthetic inducers in *Helicoverpa zea*, J. Chem. Ecol. 35 (2009) 183–189.
[128] R.S. Zeng, Z. Wen, G. Niu, M.A. Schuler, M.R. Berenbaum, Enhanced toxicity and induction of cytochrome P450s suggest a cost of “eavesdropping” in a multitrophic interaction, J. Chem. Ecol. 5 (2009) 526–532.
[129] X. Zhou, C. Ma, M. Li, C. Sheng, H. Liu, X. Qiu, CYP9A12 and CYP9A17 in the cotton bollworm, *Helicoverpa armigera*: sequence similarity, expression profile and xenobiotic response, Pestic. Manage. Sci. 66 (2010) 65–73.
[130] M.J. Snyder, J.L. Stevens, J.F. Andersen, R. Feyereisen, Expression of cytochrome P450 genes of the CYP4 family in midgut and fat body of the tobacco hornworm, *Manduca sexta*, Arch. Biochem. Biophys. 321 (1995) 13–20.
[131] J.L. Stevens, M.J. Snyder, J.F. Koener, R. Feyereisen, Inducible P450s of the CYP9 family from larval *Manduca sexta* midgut, Insect Biochem. Mol. Biol. 30 (2000) 559–568.
[132] S. Bhaskara, E.D. Dean, V. Lam, R. Ganguly, Induction of two cytochrome P450 genes, Cyp6a2 and Cyp6a8, of *Drosophila melanogaster* by caffeine in adult flies and in cell culture, Gene 377 (2006) 56–64.

[133] L. Willoughby, H. Chung, C. Lumb, C. Robin, P. Batterham, P.J. Daborn, A comparison of *Drosophila melanogaster* detoxification gene induction responses for six insecticides, caffeine and phenobarbital, Insect Biochem. Mol. Biol. 36 (2006) 934–942.
[134] L. Willoughby, P. Batterham, P.J. Daborn, Piperonyl butoxide induces the expression of cytochrome P450 and glutathione S-transferase genes in *Drosophila melanogaster*, Pestic. Manag. Sci. 63 (2007) 803–808.
[135] J.P. David, S. Boyer, A. Mesneau, A. Ball, H. Ranson, C. Dauphin-Villemant, Involvement of cytochrome P450 monooxygenases in the response of mosquito larvae to dietary plant xenobiotics, Insect Biochem. Mol. Biol. 36 (2006) 410–420.
[136] R.M. Johnson, Toxicogenomics of *Apis millifera*, Ph.D. dissertation, University of Illinois Urbana-Champaign, 2008.
[137] R. Poupardin, S. Reynaud, C. Strode, H. Ranson, J. Vontas, J.P. David, Cross-induction of detoxification genes by environmental xenobiotics and insecticides in the mosquito *Aedes aegypti*: impact on larval tolerance to chemical insecticides, Insect Biochem. Mol. Biol. 38 (2008) 540–551.
[138] R. Poupardin, M.A. Riaz, J. Vontas, J.P. David, S. Reynaud, Transcription profiling of eleven cytochrome P450s potentially involved in xenobiotic metabolism in the mosquito *Aedes aegypti*, Insect Mol. Biol. 2 (2010) 185–193.
[139] A. Brandt, M. Scharf, J.H.F. Pedra, G. Holmes, A. Dean, M. Kreitman, B.R. Pittendrigh, Differential expression and induction of two *Drosophila* cytochrome P450 genes near the *Rst(2)DDT* locus, Insect Mol. Biol. 11 (2002) 337–341.
[140] F. Zhu, T. Li, L. Zhang, N. Liu, Co-up-regulation of three P450 genes in response to permethrin exposure in permethrin resistant house flies, *Musca domestica*, BMC Physiol. 8 (2008) 18.
[141] E. Hodgson, P.E. Levi, Interactions of piperonyl butoxide with cytochrome P450, in: D.G. Jones (Ed.), Piperonyl Butoxide, Academic Press, London, 1998, pp. 41–54.
[142] H.R. Jensen, I.M. Scott, S. Sims, V.L. Trudeau, J.T. Arnason, The effect of a synergistic concentration of a *Piper nigrum* extract used in conjunction with pyrethrum upon gene expression in *Drosophila melanogaster*, Insect Mol. Biol. 15 (2006) 329–339.
[143] J. Baudry, W. Li, L. Pan, M.R. Berenbaum, M.A. Schuler, Molecular docking of substrates and inhibitors in the catalytic site of CYP6B1, an insect cytochrome P450 monooxygenase, Protein Eng. 16 (2003) 577–587.
[144] Z. Wen, M.R. Berenbaum, M.A. Schuler, Inhibition of CYP6B1-mediated detoxification of xanthotoxin by plant allelochemicals in the black swallowtail (*Papilio polyxenes*), J. Chem. Ecol. 32 (2006) 507–522.